Mohammad Taha **Askari**

PhD Candidate · Electrical and Computer Engineering, The University of British Columbia

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Education

PhD in Electrical & Computer Engineering *Vancouver, Canada*

Electrical and Computer Engineering Department, The University of British Columbia (UBC) *Sep. 2022 - May. 2027*

- Advisor: Prof. Lutz Lampe

M.A.Sc. in Electrical & Computer Engineering *Vancouver, Canada*

Electrical and Computer Engineering Department, The University of British Columbia (UBC) *Sep. 2020 - Aug. 2022*

- Cumulative GPA: 93.7 out of 100 (4.0/4.0) via 18 credits
- Advisor: Prof. Lutz Lampe

B.Sc. in Electrical Engineering *Tehran, Iran*

Department of Electical Enginerring, Sharif University of Technology (SUT) *Sep. 2015 - Jul. 2020*

- Cumulative GPA: 18.42 out of 20 (4.0/4.0) via 146 credits
- Advisor: Prof. Hamid Behroozi

Research Interests

Machine Learning for Sequential and Structured Data
Probabilistic Modeling and Joint Distribution Learning
Neural Sequence Models (Autoregressive Models, Transformers)
End-to-End Learning and Optimization
Information Theory and Statistical Inference

Research Experience

M.A.Sc. & PhD Thesis at Data Communications Group *ECE Department, UBC*

Research Assistant (Supervisor: Prof. Lutz Lampe) *Sep. 2020 - Apr. 2027*

- Conducted independent research on **probabilistic modeling and sequence design** for communication systems with memory and nonlinear dynamics.
- Developed an **analytical low-pass filter model** to characterize interactions between probabilistic shaping and nonlinear optical fiber channels, enabling interpretable system-level optimization.
- Proposed a **sequence selection algorithm** that improves robustness to nonlinear interference by exploiting temporal dependencies.
- Designed a **Bayesian carrier phase recovery algorithm** incorporating prior distributions, achieving improved performance in low-SNR regime.
- Investigated **end-to-end learning approaches** using sequence-based autoencoder architectures for probabilistic shaping and joint distribution learning.

Research Internship at Nokia Bell Labs *Greater Paris Metropolitan Region, France*

Machine Learning Research Intern *April. 2025 - April. 2026*

- Designed and implemented **Neural Probabilistic Shaping (NPS)**, an autoregressive **sequence modeling framework** using neural sequence models and **Gumbel-Softmax** sampling for joint distribution learning.
- Built an **end-to-end differentiable training pipeline** integrating neural models with differentiable channel models, enabling gradient-based optimization over sequences.
- Evaluated the proposed models on **multi-dimensional settings**, demonstrating improved robustness and scalability compared to heuristic and sampling-based methods.

Research Internship at Roche Canada *Greater Toronto Area, Canada*

Algorithm R&D Software Engineering Intern *Jun. 2024 - Oct. 2024*

- Designed and implemented **ML pipelines** for feature extraction, dataset generation, and neural network training for large-scale sequencing data.
- Developed and optimized **weakly supervised deep learning models**, including a custom **multi-label architecture** for multi-class anomaly detection and segmentation.
- Integrated **model interpretability, performance analysis, and computational complexity evaluation** into the training workflow to support iterative model improvement.

Publications and Posters

- M.T. Askari, L. Lampe, and A. Ghazisaeidi ”Neural Probabilistic Amplitude Shaping for Nonlinear Fiber Channels,” 2026 Optical Fiber Communications Conference and Exhibition (OFC).
- M.T. Askari, L. Lampe, and A. Ghazisaeidi ”Neural Probabilistic Shaping: Joint Distribution Learning for Optical Fiber Communications,” 2025 European Conference on Optical Communication (ECOC). [Online Access]
- M.T. Askari and L. Lampe, ”Probabilistic Shaping for Nonlinearity Tolerance,” Journal of Lightwave Technology. [Online Access]
- M.T. Askari and L. Lampe, ”Perturbation-based Sequence Selection for Probabilistic Amplitude Shaping,” 2024 European Conference on Optical Communication (ECOC). [Online Access]
- M.T. Askari and L. Lampe, ”Bayesian Phase Search for Probabilistic Amplitude Shaping,” 2023 European Conference on Optical Communication (ECOC). [Online Access]
- M.T. Askari, ”Interplay between Fiber Nonlinearity and Probabilistic Amplitude Shaping,” Master’s Thesis, UBC [Online Access]
- M.T. Askari, L. Lampe, and J. Mitra, ”Probabilistic Amplitude Shaping and Nonlinearity Tolerance: Analysis and Sequence Selection Method,” Journal of Lightwave Technology. [Online Access]
- M.T. Askari, L. Lampe, and J. Mitra, ”Nonlinearity Tolerant Shaping with Sequence Selection,” 2022 European Conference on Optical Communication (ECOC) [Online Access]
- M.T. Askari, ”Nonlinearity Tolerant Sequence Selection,” Poster at 17^{th} Canadian Workshop on Information Theory (CWIT)

Language Skills

English Advanced
Persian Native

Relevant Courses

CPSC 436N-Natural Language Processing	*95.0 out of 100*	**Dr. Shwartz**	*UBC*	*Winter1 2023*
EECE 571D-Detection, Estimation, and Learning	*98.0 out of 100*	**Dr. Lampe**	*UBC*	*Winter2 2021*
EECE 562-Statistical Signal Processing	*90.0 out of 100*	**Dr. Wang**	*UBC*	*Winter1 2021*
CPSC 540-Advanced Machine Learning	*96.0 out of 100*	**Dr. Schmidt**	*UBC*	*Winter2 2020*
CPSC 340-Machine Learning & Data Mining	*100 out of 100*	**Dr. Wood**	*UBC*	*Winter1 2020*
EECE 565-Communication Networks	*95.0 out of 100*	**Dr. Wong**	*UBC*	*Winter1 2020*
Introduction to Machine Learning	*18.0 out of 20*	**Dr. Mohammadzade**	*SUT*	*Spring 2019*
Information & Coding Theory	*18.7 out of 20*	**Dr. Mirmohseni**	*SUT*	*Fall 2018*

Software Proficiency

Programming Python, C/C++, Git, SQL, Matlab, Julia, R

Teaching Experience

Teaching Assistant	ELEC 221-Signals & Systems	**Dr. Yan**	*UBC*	*Winter2 2025*
Teaching Assistant	STAT 251-Elementary Statistics	**Dr. Premarathna**	*UBC*	*Winter2 2025*
Teaching Assistant	STAT 251-Elementary Statistics	**Dr. Premarathna**	*UBC*	*Winter2 2024*
Teaching Assistant	STAT 200-Elementary Statistics for Applications	**Dr. Lourenzutti**	*UBC*	*Winter1 2023*
Teaching Assistant	ELEC 221-Signals & Systems	**Dr. Thrampoulidis**	*UBC*	*Winter1 2023*
Teaching Assistant	STAT 200-Elementary Statistics for Applications	**Dr. Burr**	*UBC*	*Summer1 2023*
Teaching Assistant	STAT 251-Elementary Statistics	**Dr. Premarathna**	*UBC*	*Winter2 2022*
Teaching Assistant	ELEC 221-Signals & Systems	**Dr. Di Matteo**	*UBC*	*Winter1 2022*
Teaching Assistant	STAT 200-Elementary Statistics for Applications	**Dr. Burr**	*UBC*	*Winter1 2022*
Teaching Assistant	CPSC 330-Applied Machine Learning	**Dr. Oveisi**	*UBC*	*Summer1 2022*
Teaching Assistant	STAT 306-Finding Relationships in Data	**Dr. Premarathna**	*UBC*	*Winter1 2021*
Teaching Assistant	CPSC 303-Numerical Approximation	**Dr. Yi**	*UBC*	*Winter2 2020*

Voluntary Experience

ECE Graduate Student Association (ECEGSA) *ECE Department, UBC*

VP ACADEMIC *Apr. 2024-Apr. 2025*

- Organized academic and professional development events, including resume workshops, career fairs, and lightning talks, to support students' career advancement and academic success.
- Facilitated opportunities for graduate students to build their professional brand and network with industry professionals and academic peers.

Online Instructional Skills Workshop *CIRTL, UBC*

PARTICIPANT *Jan. 2021*

- Practiced developing online lesson plans and using techniques to involve students actively in the online learning process.
- Taught three lessons to employ new teaching techniques.

Conference on "Emerging Technologies for Electric Vehicles and Smart Cars" *EE Department, SUT*

ADMINISTRATIVE HEAD *Nov. 2018*

- Directed a team of ten students in holding the conference.
- Gained knowledge and experience in several managerial skills like apportioning responsibilities and group management.

International Biology Olympiad *Tehran, Iran*

TEAM GUIDE *Jul. 2018*

- Guided Czech Republic's delegation in the 29th International Biology Olympiad (IBO).
- Acquired experience in the communication skills required for accompanying the delegation.
- Awarded certification from the IBO committee.

Honors & Awards

Awarded UBC 2025 PhD Graduate Student Initiative (GSI) with an amount of $8,200. *2025*

Nominated for UBC Four Year Doctoral Fellowship with an amount of $18,200 per year. *2023*

Awarded UBC 2021 M.A.Sc. Graduate Student Initiative (GSI) with an amount of $5,000. *2021*

Member of the National Elites Foundation by being in the best ranks of the University entrance exam. *2015-Present*

Selected Course Projects

COVID-19 Detection: A Combination of Transformer Models with Image Augmentation Methods *Winter 2020*

COURSE: ADVANCED MACHINE LEARNING *UBC*

- Studied methods for generating synthetic X-ray images to improve the accuracy of COVID-19 detection models and implemented GAN and VAE to augment data.
- Applied transformer models to COVID-19 classification based on chest X-ray images.
- Supervisor: Dr. Schmidt

Trajectory Prediction *Winter 2020*

COURSE: MACHINE LEARNING & DATA MINING *UBC*

- Studied methods for autonomous driving trajectory prediction and implemented a chained neural network model.
- Supervisor: Dr. Wood